

Full Molecular Hamiltonian

From Classical to Quantum: the exact non-relativistic quantization

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July 22, 2026

Roadmap: four stages

We build the **exact** non-relativistic molecular Hamiltonian in four stages:

1. **Classical** — lab-frame Hamiltonian; transformation into center-of-mass coordinates produces the mass polarization (mass pol.).
2. **Body-Fixed** — co-rotating frame + Lagrangian $\Rightarrow$ classical internal molecular Hamiltonian H_{int} .
3. **Quantize** — **Podolsky quantization** for curvilinear coordinates in polyatomic molecules.
4. **Quantum** $\hat{H}$ — the exact non-relativistic molecular Hamiltonian for non-linear polyatomic molecules.

Outline

Motivation

Classical Cartesian Hamiltonian

Mass-centered coordinates

Body-fixed frame and the Lagrangian

The quantization problem

The ro-vibrational geometry

Quantizing rotation & Coriolis coupling

The vibrational block & the Watson pseudopotential

The complete quantum Hamiltonian

Summary

Motivation

Why is the molecular Hamiltonian non-trivial?

The lab Hamiltonian couples *all* particles through the Coulomb potential:

$$H = \sum_j \frac{\mathbf{P}_j^2}{2M_j} + \sum_i \frac{\mathbf{p}_i^2}{2m_e} + V.$$

To do molecular physics we want to separate

$$\underbrace{\text{translation}}_{\text{COM}} \quad | \quad \underbrace{\text{rotation}}_{\text{Euler angles } \Omega} \quad | \quad \underbrace{\text{vibration}}_{\text{normal coords } Q} \quad | \quad \underbrace{\text{electrons}}_{\vec{r}_i}.$$

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The catch

Rotation + vibration are *curvilinear* coordinates. Naively quantizing $p \rightarrow -i\hbar \partial$ is **not** Hermitian there. Getting it right forces the Podolsky quantization.

What we will not approximate

We keep everything exact until the very end:

- **No** Born–Oppenheimer / clamped nuclei.
- **No** neglect of *mass polarization*.
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The *only* ingredients are:

1. non-relativistic dynamics;
2. the Eckart frame (a *convention*, not an approximation);
3. parametrizing the nuclear shape by $3N_{\text{nuc}} - 6$ normal coordinates (non-linear molecules).

The endpoint is the exact quantum-mechanical internal molecular Hamiltonian $\hat{H}_{\text{int}}$ — the rigorous starting point

Classical Cartesian Hamiltonian

The Lagrangian in the lab frame

N_{nuc} nuclei $(M_j, Z_j e, \mathbf{R}_j)$ and N_{elec} electrons $(m_e, -e, \mathbf{r}_i)$, all in the space-fixed frame; SI units.
The Lagrangian is kinetic minus potential:

$$\mathcal{L} = \sum_j \frac{1}{2} M_j \dot{\mathbf{R}}_j^2 + \sum_i \frac{1}{2} m_e \dot{\mathbf{r}}_i^2 - V(\mathbf{R}_j, \mathbf{r}_i).$$

Differentiate $\mathcal{L}$ with respect to each velocity to get the **canonical momenta**:

$$\mathbf{P}_j \equiv \frac{\partial \mathcal{L}}{\partial \dot{\mathbf{R}}_j} = M_j \dot{\mathbf{R}}_j, \quad \mathbf{p}_i \equiv \frac{\partial \mathcal{L}}{\partial \dot{\mathbf{r}}_i} = m_e \dot{\mathbf{r}}_i.$$

Canonical brackets $\{R_{j\alpha}, P_{k\beta}\} = \delta_{jk} \delta_{\alpha\beta}$, $\{r_{i\alpha}, p_{j\beta}\} = \delta_{ij} \delta_{\alpha\beta}$.

- Kinetic energy is diagonal and quadratic in Cartesian velocities: *canonical momentum* = *mass* $\times$ *velocity*.

Legendre transform to the Hamiltonian

The three-move recipe (we reuse it, less trivially, in a moment):

1. **Define** $p_q \equiv \partial\mathcal{L}/\partial\dot{q}$ for every coordinate.
2. **Invert** for every velocity as a function of the momenta.
3. **Assemble** $H \equiv \sum_q p_q \dot{q} - \mathcal{L}$ and substitute.

Here step 2 is immediate, $\dot{\mathbf{R}}_j = \mathbf{P}_j/M_j$, $\dot{\mathbf{r}}_i = \mathbf{p}_i/m_e$; step 3 gives

$$H = \underbrace{\sum_j \frac{\mathbf{P}_j^2}{2M_j}}_{T_N} + \underbrace{\sum_i \frac{\mathbf{p}_i^2}{2m_e}}_{T_e} + V, \quad (1.1)$$

$$V = \frac{1}{4\pi\epsilon_0} \left(\frac{1}{2} \sum_{i \neq j} \frac{Z_i Z_j e^2}{R_{ij}} - \sum_{i,j} \frac{Z_j e^2}{|\mathbf{R}_j - \mathbf{r}_i|} + \frac{1}{2} \sum_{i \neq j} \frac{e^2}{r_{ij}} \right). \quad (1.2)$$

Everything that follows is an exact rewriting of (1.1)–(1.2).

Mass-centered coordinates

Two centers of mass, two roles

The translational coordinate and the internal reference point **need not be the same**. Define

$$M_N = \sum_j M_j, \quad M_{\text{tot}} = M_N + N_{\text{elec}} m_e,$$

$$\mathbf{R}_{\text{ncm}} = \frac{1}{M_N} \sum_j M_j \mathbf{R}_j, \quad \mathbf{R}_{\text{cm}} = \frac{1}{M_{\text{tot}}} \left(\sum_j M_j \mathbf{R}_j + m_e \sum_i \mathbf{r}_i \right), \quad (\text{A.2, A.3})$$

and refer every internal coordinate — nuclear and electronic — to the *nuclear* CoM:

$$\mathbf{R}'_j = \mathbf{R}_j - \mathbf{R}_{\text{ncm}}, \quad \mathbf{r}'_i = \mathbf{r}_i - \mathbf{R}_{\text{ncm}}, \quad \sum_j M_j \mathbf{R}'_j = 0. \quad (\text{A.4})$$

The mixed choice

Translation = total CoM $\Rightarrow$ correct total mass, and (we will see) the COM–internal cross terms cancel identically.

Internals referred to the nuclear CoM $\Rightarrow$ Born–Oppenheimer-friendly coordinates, V form-invariant, and the mass polarization appears *by itself*.

Inverting: every particle rides a shared drift

Solve (A.3) for the nuclear CoM, then invert (A.4) for the lab positions:

$$\mathbf{R}_{\text{ncm}} = \mathbf{R}_{\text{cm}} - \frac{m_e}{M_{\text{tot}}} \sum_l \mathbf{r}'_l, \quad (\text{A.5})$$

$$\mathbf{R}_j = \mathbf{R}_{\text{cm}} + \mathbf{R}'_j - \frac{m_e}{M_{\text{tot}}} \sum_l \mathbf{r}'_l, \quad \mathbf{r}_i = \mathbf{R}_{\text{cm}} + \mathbf{r}'_i - \frac{m_e}{M_{\text{tot}}} \sum_l \mathbf{r}'_l. \quad (\text{A.6})$$

Both families ride on the *same* drift — the nuclear-CoM velocity:

$$\boxed{\mathbf{D} \equiv \dot{\mathbf{R}}_{\text{ncm}} = \dot{\mathbf{R}}_{\text{cm}} - \frac{m_e}{M_{\text{tot}}} \sum_l \dot{\mathbf{r}}'_l, \quad \dot{\mathbf{R}}_j = \mathbf{D} + \dot{\mathbf{R}}'_j, \quad \dot{\mathbf{r}}_i = \mathbf{D} + \dot{\mathbf{r}}'_i.} \quad (\text{A.7})$$

Every lab velocity = internal velocity + one shared drift $\mathbf{D}$ — just like passengers on a train.

Kinetic energy in velocity form: the mass polarization appears

Substitute (A.7) into $T = \sum_j \frac{1}{2} M_j \dot{\mathbf{R}}_j^2 + \sum_i \frac{1}{2} m_e \dot{\mathbf{r}}_i^2$. The nuclear cross term dies by $\sum_j M_j \dot{\mathbf{R}}'_j = 0$; the electronic one does *not*:

$$T_N = \frac{1}{2} M_N \mathbf{D}^2 + \frac{1}{2} \sum_j M_j \dot{\mathbf{R}}_j'^2, \quad T_e = \frac{1}{2} N_{\text{elec}} m_e \mathbf{D}^2 + m_e \mathbf{D} \cdot \sum_i \dot{\mathbf{r}}'_i + \frac{1}{2} m_e \sum_i \dot{\mathbf{r}}_i'^2. \quad (\text{A.8, A.9})$$

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Expand $\mathbf{D}$ in the two $\mathbf{D}$ -dependent pieces (A.11a,b): the $\dot{\mathbf{R}}_{\text{cm}} \cdot \sum \dot{\mathbf{r}}'$ cross terms cancel **identically**, and the quadratic pieces combine to one *negative* remainder:

$$T = \frac{1}{2} M_{\text{tot}} \dot{\mathbf{R}}_{\text{cm}}^2 + \frac{1}{2} \sum_j M_j \dot{\mathbf{R}}_j'^2 + \frac{1}{2} m_e \sum_i \dot{\mathbf{r}}_i'^2 - \frac{m_e^2}{2M_{\text{tot}}} \left(\sum_i \dot{\mathbf{r}}'_i \right)^2. \quad (\text{A.12})$$

Canonical momenta: $\partial\mathcal{L}/\partial\dot{q}$, and the tricky one

With $\mathcal{L} = T - V$ (V velocity-independent), differentiate the boxed T :

$$\mathbf{P}_{\text{cm}} = \frac{\partial\mathcal{L}}{\partial\dot{\mathbf{R}}_{\text{cm}}} = M_{\text{tot}}\dot{\mathbf{R}}_{\text{cm}}, \quad \mathbf{P}'_j = \frac{\partial\mathcal{L}}{\partial\dot{\mathbf{R}}'_j} = M_j\dot{\mathbf{R}}'_j. \quad (\text{A.13, A.17})$$

The electrons take care: *two* terms of T involve $\dot{\mathbf{r}}'_i$ (its own kinetic term, *and* the mass-polarization sum), so

$$\boxed{\mathbf{p}'_i \equiv \frac{\partial\mathcal{L}}{\partial\dot{\mathbf{r}}'_i} = m_e\dot{\mathbf{r}}'_i - \frac{m_e^2}{M_{\text{tot}}} \sum_l \dot{\mathbf{r}}'_l \neq m_e\dot{\mathbf{r}}'_i.} \quad (\text{A.14})$$

Inverting the electron relation

Sum (A.14) over all electrons; with $M_{\text{tot}} - N_{\text{elec}} m_e = M_N$,

$$\sum_i \mathbf{p}'_i = m_e \left(1 - \frac{N_{\text{elec}} m_e}{M_{\text{tot}}} \right) \sum_I \dot{\mathbf{r}}'_I = \frac{m_e M_N}{M_{\text{tot}}} \sum_I \dot{\mathbf{r}}'_I. \quad (\text{A.15})$$

Solve for $\dot{\mathbf{r}}'_i$:

$$\boxed{\dot{\mathbf{r}}'_i = \frac{\mathbf{p}'_i}{m_e} + \frac{1}{M_N} \sum_I \mathbf{p}'_I.} \quad (\text{A.16})$$

Legendre transform: kinetic energy in momentum form

CoM and nuclear terms are immediate: $\frac{1}{2} M_{\text{tot}} \dot{\mathbf{R}}_{\text{cm}}^2 = \mathbf{P}_{\text{cm}}^2 / 2M_{\text{tot}}$, $\frac{1}{2} M_j \dot{\mathbf{R}}_j'^2 = \mathbf{P}_j'^2 / 2M_j$.
 Substituting (A.16) into the electron sector (A.19, A.20) and collecting the $(\sum_i \mathbf{p}_i')^2$ coefficients,

$$\frac{1}{M_N} + \frac{N_{\text{elec}} m_e - M_{\text{tot}}}{2M_N^2} = \frac{1}{M_N} - \frac{M_N}{2M_N^2} = +\frac{1}{2M_N}. \quad (\text{A.21})$$

As a result, the overall kinetic energy in momentum form is

$$T = \frac{\mathbf{P}_{\text{cm}}^2}{2M_{\text{tot}}} + \sum_j \frac{\mathbf{P}_j'^2}{2M_j} + \sum_i \frac{\mathbf{p}_i'^2}{2m_e} + \frac{1}{2M_N} \left(\sum_i \mathbf{p}_i' \right)^2. \quad (\text{A.22})$$

The Hamiltonian — and why the sign flips

Both internal families share the reference point $\mathbf{R}_{\text{ncm}}$, so $V(\mathbf{R}'_j, \mathbf{r}'_i)$ is unchanged in form. Add it to (A.22):

$$H = \frac{\mathbf{P}_{\text{cm}}^2}{2M_{\text{tot}}} + \underbrace{\sum_j \frac{\mathbf{P}'_j{}^2}{2M_j} + \sum_i \frac{\mathbf{p}'_i{}^2}{2m_e} + \frac{1}{2M_N} \left(\sum_i \mathbf{p}'_i \right)^2}_{H_{\text{int}} \text{ (2.13)}} + V(\mathbf{R}'_j, \mathbf{r}'_i). \quad (2.12 \equiv \text{A.23})$$

Body-fixed frame and the Lagrangian

Body-fixed frame: rotate with the nuclei

Introduce a rotation $\mathbf{S}(\Omega)$ (Euler angles $\Omega = \phi, \theta, \chi$) and body-fixed (BF) coordinates:

$$\mathbf{R}'_j = \mathbf{S}(\Omega) \bar{\mathbf{R}}_j(Q), \quad \mathbf{r}'_i = \mathbf{S}(\Omega) \bar{\mathbf{r}}_i. \quad (3.1)$$

The *Eckart* conditions fix $\mathbf{S}$ from the *nuclei*; the nuclear shape is rectilinear mass-weighted **normal coordinates** Q_b :

$$\bar{\mathbf{R}}_j(Q) = \bar{\mathbf{R}}_j^{\text{eq}} + \frac{1}{\sqrt{M_j}} \sum_b \mathbf{I}_{jb} Q_b, \quad \sum_j \mathbf{I}_{ja} \cdot \mathbf{I}_{jb} = \delta_{ab}. \quad (3.1a)$$

The BF angular velocity $\boldsymbol{\omega}$ is defined by

$$(\mathbf{S}^T \dot{\mathbf{S}}) \mathbf{v} = \boldsymbol{\omega} \times \mathbf{v}$$

- Try to differentiate $\mathbf{S}^T \mathbf{S} = 0$, you will get $\mathbf{S}^T \dot{\mathbf{S}}$ is antisymmetric, so it can be written as a cross product with some vector $\boldsymbol{\omega}$.

The velocity-form Lagrangian — everything, electrons included

Substitute the co-rotating velocities (3.3) $\dot{\mathbf{R}}'_j = \mathbf{S}(\boldsymbol{\omega} \times \bar{\mathbf{R}}_j + \dot{\bar{\mathbf{R}}}_j)$, $\dot{\mathbf{r}}'_i = \mathbf{S}(\boldsymbol{\omega} \times \bar{\mathbf{r}}_i + \dot{\bar{\mathbf{r}}}_i)$ to Lagrangian (A.12) and use $\mathbf{S}^T \mathbf{S} = \mathbb{1}$ to get the BF velocity form:

$$\mathcal{L} = \frac{1}{2} M_{\text{tot}} |\dot{\mathbf{R}}_{\text{cm}}|^2 + \frac{1}{2} \sum_j M_j |\boldsymbol{\omega} \times \bar{\mathbf{R}}_j + \dot{\bar{\mathbf{R}}}_j|^2 + \frac{m_e}{2} \sum_i |\boldsymbol{\omega} \times \bar{\mathbf{r}}_i + \dot{\bar{\mathbf{r}}}_i|^2 - \frac{m_e^2}{2M_{\text{tot}}} |\boldsymbol{\omega} \times \bar{\mathbf{r}} + \dot{\bar{\mathbf{r}}}|^2 - V \quad (3.7')$$

with $\bar{\mathbf{r}} \equiv \sum_i \bar{\mathbf{r}}_i$. The potential is now manifestly $\mathbf{S}$ -independent.

The vibrational term: $\dot{\mathbf{R}}_j \rightarrow \dot{Q}_b$

Expanding each square $|\boldsymbol{\omega} \times \mathbf{a} + \dot{\mathbf{a}}|^2$ leaves a pure-velocity (**vibrational**) piece $|\dot{\mathbf{a}}|^2$. For the nuclei it collapses cleanly: the mode vectors are *constant* [$\partial \bar{\mathbf{R}}_j / \partial Q_b = \mathbf{l}_{jb} / \sqrt{M_j}$ from (3.1a)], so

$$\dot{\mathbf{R}}_j = \frac{1}{\sqrt{M_j}} \sum_b \mathbf{l}_{jb} \dot{Q}_b.$$

The M_j cancels and the orthonormality $\sum_j \mathbf{l}_{jb} \cdot \mathbf{l}_{jc} = \delta_{bc}$ (3.1a) collapses the double sum:

$$T_{\text{vib}}^N \equiv \frac{1}{2} \sum_j M_j |\dot{\mathbf{R}}_j|^2 = \frac{1}{2} \sum_{b,c} \underbrace{\left(\sum_j \mathbf{l}_{jb} \cdot \mathbf{l}_{jc} \right)}_{=\delta_{bc}} \dot{Q}_b \dot{Q}_c = \frac{1}{2} \sum_b \dot{Q}_b^2.$$

Expand every squared term $\Rightarrow$ the full inertia

The vector identity (any $\mathbf{a}, \dot{\mathbf{a}}, \boldsymbol{\omega}$)

$$\frac{m}{2} |\boldsymbol{\omega} \times \mathbf{a} + \dot{\mathbf{a}}|^2 = \underbrace{\frac{m}{2} \boldsymbol{\omega}^T (a^2 \mathbb{1} - \mathbf{a} \mathbf{a}^T) \boldsymbol{\omega}}_{\text{rotational}} + \underbrace{m \boldsymbol{\omega} \cdot (\mathbf{a} \times \dot{\mathbf{a}})}_{\text{Coriolis}} + \underbrace{\frac{m}{2} |\dot{\mathbf{a}}|^2}_{\text{vibrational}} \quad (3.9c)$$

applied to *nuclei, electrons, and the mass-polarization term* gives

$$\mathcal{L} = \frac{1}{2} M_{\text{tot}} |\dot{\mathbf{R}}_{\text{cm}}|^2 + \frac{1}{2} \boldsymbol{\omega}^T \underbrace{(\mathbf{I}_N + \mathbf{I}_e - \mathbf{I}_X)}_{\text{full inertia } \mathbf{I}} \boldsymbol{\omega} + \boldsymbol{\omega} \cdot (\mathbf{L}_N^{\text{vib}} + \mathbf{L}_e - \mathbf{L}_X) + \frac{1}{2} \sum_b \dot{Q}_b^2 + T_e^{\text{BF}} - T_X^{\text{mp}} - V \quad (3.13')$$

Definitions

$$\text{Inertia } \mathbf{I}_N = \sum_j M_j (\bar{R}_j^2 \mathbb{1} - \bar{\mathbf{R}}_j \bar{\mathbf{R}}_j^T), \quad \mathbf{I}_e = m_e \sum_i (\bar{r}_i^2 \mathbb{1} - \bar{\mathbf{r}}_i \bar{\mathbf{r}}_i^T), \quad \mathbf{I}_X = \frac{m_e^2}{M_{\text{tot}}} (\bar{r}^2 \mathbb{1} - \bar{\mathbf{r}} \bar{\mathbf{r}}^T)$$

$$\text{Ang. mom. } \mathbf{L}_N^{\text{vib}} = \sum_j M_j \bar{\mathbf{R}}_j \times \dot{\bar{\mathbf{R}}}_j, \quad \mathbf{L}_e = m_e \sum_i \bar{\mathbf{r}}_i \times \dot{\bar{\mathbf{r}}}_i, \quad \mathbf{L}_X = \frac{m_e^2}{M_{\text{tot}}} \bar{\mathbf{r}} \times \dot{\bar{\mathbf{r}}}$$

$$\text{Kinetic } T_e^{\text{BF}} = \frac{m_e}{2} \sum_i |\dot{\bar{\mathbf{r}}}_i|^2, \quad T_X^{\text{mp}} = \frac{m_e^2}{2M_{\text{tot}}} |\dot{\bar{\mathbf{r}}}|^2, \quad T_{\text{vib}}^N = \frac{1}{2} \sum_b \dot{Q}_b^2$$

Total, nuclear, and canonical electronic angular momenta

Read the momenta off $\mathcal{L}$ (3.13'):

$$\mathbf{J} \equiv \frac{\partial \mathcal{L}}{\partial \boldsymbol{\omega}} = \underbrace{(\mathbf{I}_N + \mathbf{I}_e - \mathbf{I}_X) \boldsymbol{\omega} + \mathbf{L}_N^{\text{vib}} + \mathbf{L}_e - \mathbf{L}_X}_{\text{total ang. mom.}}, \quad \mathbf{J}_N \equiv \mathbf{I}_N \boldsymbol{\omega} + \mathbf{L}_N^{\text{vib}} \quad (\text{nuclear}). \quad (3.16, 3.15)$$

The **canonical electronic angular momentum** is built from the electron pair $(\bar{\mathbf{r}}_i, \bar{\mathbf{p}}_i)$:

$$\mathbf{L}_{\text{elec}} = \sum_i \bar{\mathbf{r}}_i \times \bar{\mathbf{p}}_i, \quad \bar{\mathbf{p}}_i \equiv \frac{\partial \mathcal{L}}{\partial \dot{\bar{\mathbf{r}}}_i} = m_e (\boldsymbol{\omega} \times \bar{\mathbf{r}}_i + \dot{\bar{\mathbf{r}}}_i) - \frac{m_e^2}{M_{\text{tot}}} (\boldsymbol{\omega} \times \bar{\mathbf{r}} + \dot{\bar{\mathbf{r}}}). \quad (3.14)$$

Vibrational momentum, with the **Coriolis vectors** $\boldsymbol{\sigma}_b$:

$$P_b = \frac{\partial \mathcal{L}}{\partial \dot{Q}_b} = \boldsymbol{\omega} \cdot \boldsymbol{\sigma}_b + \dot{Q}_b, \quad \boldsymbol{\sigma}_b = \sum_a \zeta_{ab} Q_a, \quad \zeta_{ab} = \sum_j \mathbf{l}_{ja} \times \mathbf{l}_{jb}. \quad (3.17a,b)$$

Proof 1: $\mathbf{L}_{\text{elec}}$ is the electron + mass-polarization ang. mom.

Substitute $\bar{\mathbf{p}}_i$ (3.14) into $\mathbf{L}_{\text{elec}} = \sum_i \bar{\mathbf{r}}_i \times \bar{\mathbf{p}}_i$ and use BAC-CAB on the two $\boldsymbol{\omega}$ -terms,

$$\mathbf{a} \times (\boldsymbol{\omega} \times \mathbf{a}) = (a^2 \mathbb{1} - \mathbf{a} \mathbf{a}^T) \boldsymbol{\omega} :$$

$$\boxed{\mathbf{L}_{\text{elec}} = (\mathbf{I}_e - \mathbf{I}_X) \boldsymbol{\omega} + (\mathbf{L}_e - \mathbf{L}_X).} \quad (3.18)$$

What this says

The canonical electronic angular momentum is *exactly* the electron and mass-polarization inertia/Coriolis pieces of $\mathcal{L}$ (3.13'). Therefore the total splits cleanly,

$$\mathbf{J} = \underbrace{\mathbf{I}_N \boldsymbol{\omega} + \mathbf{L}_N^{\text{vib}}}_{\mathbf{J}_N} + \underbrace{(\mathbf{I}_e - \mathbf{I}_X) \boldsymbol{\omega} + (\mathbf{L}_e - \mathbf{L}_X)}_{\mathbf{L}_{\text{elec}}} = \mathbf{J}_N + \mathbf{L}_{\text{elec}}. \quad (3.16)$$

Proof 2: the electron sector

Write the body-frame electron velocity $\mathbf{w}_i \equiv \boldsymbol{\omega} \times \bar{\mathbf{r}}_i + \dot{\bar{\mathbf{r}}}_i$. The electron + mass-polarization part of $\mathcal{L}$ (3.7') is

$$\mathcal{L}_e = \frac{m_e}{2} \sum_i |\mathbf{w}_i|^2 - \frac{m_e^2}{2M_{\text{tot}}} \left| \sum_i \mathbf{w}_i \right|^2, \quad \bar{\mathbf{p}}_i = m_e \mathbf{w}_i - \frac{m_e^2}{M_{\text{tot}}} \sum_l \mathbf{w}_l,$$

identical in structure to the Sec. 2 velocity form (A.12) with $\dot{\mathbf{r}}'_i \rightarrow \mathbf{w}_i$. The same mass-matrix inversion $\mathbf{A} \rightarrow \mathbf{A}^{-1}$ (A.25–A.26) gives

$$\sum_i \bar{\mathbf{p}}_i \cdot \mathbf{w}_i - \mathcal{L}_e = \sum_i \frac{\bar{\mathbf{p}}_i^2}{2m_e} + \frac{1}{2M_N} \left(\sum_i \bar{\mathbf{p}}_i \right)^2$$

The key identity

From the full angular momentum $\mathbf{J} = \partial\mathcal{L}/\partial\boldsymbol{\omega} = (\mathbf{I}_N + \mathbf{I}_e - \mathbf{I}_X)\boldsymbol{\omega} + \mathbf{L}_N^{\text{vib}} + \mathbf{L}_e - \mathbf{L}_X$ and the electronic one $\mathbf{L}_{\text{elec}} = (\mathbf{I}_e - \mathbf{I}_X)\boldsymbol{\omega} + (\mathbf{L}_e - \mathbf{L}_X)$ (Proof 1, 3.18), subtract: the electronic and mass-polarization pieces **cancel exactly**, leaving a purely *nuclear* right side — rotational kernel $\mathbf{I}_N$ alone, all electronic content carried by the shift $\mathbf{J} \rightarrow \mathbf{J} - \mathbf{L}_{\text{elec}}$:

$$\boxed{\mathbf{J} - \mathbf{L}_{\text{elec}} = \mathbf{I}_N(Q) \boldsymbol{\omega} + \mathbf{L}_N^{\text{vib}}.} \quad (3.19)$$

To invert (3.19) for $\boldsymbol{\omega}$ we must write $\mathbf{L}_N^{\text{vib}}$ in canonical variables — first the **Coriolis vector** explicitly. Using $\partial\bar{\mathbf{R}}_j/\partial Q_b = \mathbf{I}_{jb}/\sqrt{M_j}$ (3.1a) and $\bar{\mathbf{R}}_j = \bar{\mathbf{R}}_j^{\text{eq}} + \frac{1}{\sqrt{M_j}} \sum_a \mathbf{I}_{ja} Q_a$, the *rotational Eckart condition* $\sum_j \sqrt{M_j} \bar{\mathbf{R}}_j^{\text{eq}} \times \mathbf{I}_{jb} = 0$ kills the equilibrium term:

$$\boldsymbol{\sigma}_b = \sum_j M_j \bar{\mathbf{R}}_j \times \frac{\partial \bar{\mathbf{R}}_j}{\partial Q_b} = \sum_j \sqrt{M_j} \bar{\mathbf{R}}_j \times \mathbf{I}_{jb} = \sum_a \zeta_{ab} Q_a, \quad \zeta_{ab} = \sum_j \mathbf{I}_{ja} \times \mathbf{I}_{jb} \quad (\text{const}). \quad (3.17a)$$

Watson reciprocal inertia

With $\dot{\bar{\mathbf{R}}}_j = \sum_b \frac{\partial \bar{\mathbf{R}}_j}{\partial Q_b} \dot{Q}_b$, the velocity-form angular momentum is $\mathbf{L}_N^{\text{vib}} = \sum_j M_j \bar{\mathbf{R}}_j \times \dot{\bar{\mathbf{R}}}_j = \sum_b \boldsymbol{\sigma}_b \dot{Q}_b$. Now eliminate $\dot{Q}_b = P_b - \boldsymbol{\omega} \cdot \boldsymbol{\sigma}_b$ (3.17b), using $\boldsymbol{\sigma}_b(\boldsymbol{\sigma}_b \cdot \boldsymbol{\omega}) = \boldsymbol{\sigma}_b \boldsymbol{\sigma}_b^T \boldsymbol{\omega}$:

$$\mathbf{L}_N^{\text{vib}} = \sum_b \boldsymbol{\sigma}_b (P_b - \boldsymbol{\omega} \cdot \boldsymbol{\sigma}_b) = \boldsymbol{\pi} - \boldsymbol{\sigma} \boldsymbol{\sigma}^T \boldsymbol{\omega}, \quad \boldsymbol{\pi} \equiv \sum_b \boldsymbol{\sigma}_b P_b, \quad \boldsymbol{\sigma} \boldsymbol{\sigma}^T \equiv \sum_b \boldsymbol{\sigma}_b \boldsymbol{\sigma}_b^T. \quad (3.20a)$$

Substitute into the key identity (3.19), $\mathbf{J} - \mathbf{L}_{\text{elec}} = (\mathbf{I}_N - \boldsymbol{\sigma} \boldsymbol{\sigma}^T) \boldsymbol{\omega} + \boldsymbol{\pi}$, and invert for $\boldsymbol{\omega}$:

$$\boldsymbol{\omega} = \boldsymbol{\mu} (\mathbf{J} - \mathbf{L}_{\text{elec}} - \boldsymbol{\pi}), \quad \boldsymbol{\mu} \equiv (\mathbf{I}_N - \boldsymbol{\sigma} \boldsymbol{\sigma}^T)^{-1}. \quad (3.20)$$

$\boldsymbol{\mu}$ is the **Watson reciprocal-inertia tensor**, $\boldsymbol{\pi}$ the field-free vibrational angular momentum. The whole rotation–vibration coupling is now carried by $\boldsymbol{\mu}$ and the shift $\mathbf{J} \rightarrow \mathbf{J} - \mathbf{L}_{\text{elec}} - \boldsymbol{\pi}$ — *nuclear* quantities only, no Wilson G -matrix.

The exact classical molecular Hamiltonian

Legendre back. The rovibrational block gives $\frac{1}{2}(\mathbf{J} - \mathbf{L}_{\text{elec}} - \boldsymbol{\pi})^T \boldsymbol{\mu}(\mathbf{J} - \mathbf{L}_{\text{elec}} - \boldsymbol{\pi}) + \frac{1}{2} \sum_b P_b^2$; the electron sector gives H_e (Proof 2):

$$\begin{aligned}
 H_{\text{int}} = & \frac{1}{2}(\mathbf{J} - \mathbf{L}_{\text{elec}} - \boldsymbol{\pi})^T \boldsymbol{\mu}(Q)(\mathbf{J} - \mathbf{L}_{\text{elec}} - \boldsymbol{\pi}) + \frac{1}{2} \sum_b P_b^2 \\
 & + \sum_i \frac{\bar{\mathbf{p}}_i^2}{2m_e} + \frac{1}{2M_N} \left(\sum_i \bar{\mathbf{p}}_i \right)^2 + V_{NN} + V_{Ne} + V_{ee}.
 \end{aligned}
 \tag{3.22}$$

Exact, classical, in BF + Eckart coordinates.

The quantization problem

Primer: wavefunctions live in a space, L^2

In quantum mechanics a state is a **wavefunction** $\psi(q)$ — and wavefunctions add and scale like vectors, so they form a vector space. To measure “length” and “angle” we need an **inner product**; for two states φ, ψ ,

$$\langle \varphi | \psi \rangle \equiv \int \varphi^*(q) \psi(q) dq \quad (\text{Dirac “bra-ket” notation}).$$

The squared length is the **norm**, and a physical state is **normalized**:

$$\|\psi\|^2 = \langle \psi | \psi \rangle = \int |\psi(q)|^2 dq = 1 \quad (\text{total probability} = 1).$$

What is L^2 ?

L^2 is the space of **square-integrable** functions — those with finite $\int |\psi|^2 < \infty$, so the norm makes sense. Operators like $\hat{H}$ act on this space, and observables must be **Hermitian**, $\hat{O}^\dagger = \hat{O}$, to have real eigenvalues.

Primer: the measure — what you integrate against

The inner product needs a rule for “how much volume” each point carries — the **measure**. In plain Cartesian coordinates it is $dq = dx dy dz$, but after a change of variables it picks up the **Jacobian**:

$$\int f dx dy dz = \int f \underbrace{r^2 \sin \theta}_{\text{Jacobian}} dr d\theta d\phi \implies \langle \varphi | \psi \rangle = \int \varphi^* \psi \underbrace{w(q)}_{\text{measure}} d^n q.$$

Normalization is always against this measure: $\int |\psi|^2 w d^n q = 1$. We write the space as $L^2(w d^n q)$.

Why it matters here

Hermiticity depends on the measure. On a curved measure $-i\hbar\partial$ is *not* Hermitian (next slides show this for $w = r^2 \sin \theta$). Getting the operator right on the correct measure is exactly the quantization problem — and its cure is the Podolsky rule. For the molecule the measure will be $\sqrt{g} d^n q$ (later reshaped to the flat $d\mu_W$).

A worked example: one particle in spherical coordinates

Take a single free particle of mass m . Cartesian: $\mathcal{L} = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2 + \dot{z}^2)$. In spherical coordinates (r, θ, ϕ) ,

$$\mathcal{L} = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2 + r^2\sin^2\theta\dot{\phi}^2).$$

Canonical momenta $p_a = \partial\mathcal{L}/\partial\dot{q}^a$:

$$p_r = m\dot{r}, \quad p_\theta = mr^2\dot{\theta}, \quad p_\phi = mr^2\sin^2\theta\dot{\phi}.$$

Legendre transform $\Rightarrow$ the classical Hamiltonian

$$H = \frac{p_r^2}{2m} + \frac{p_\theta^2}{2mr^2} + \frac{p_\phi^2}{2mr^2\sin^2\theta}.$$

The coordinates are *curvilinear*: p_θ, p_ϕ are not mass $\times$ velocity, and the coefficients depend on r, θ .

Naive quantization gives the wrong operator

Substitute $p_a \rightarrow -i\hbar \partial_a$ into H :

$$\hat{H}_{\text{naive}} = -\frac{\hbar^2}{2m} \left[\partial_r^2 + \frac{1}{r^2} \partial_\theta^2 + \frac{1}{r^2 \sin^2 \theta} \partial_\phi^2 \right].$$

But the true kinetic operator is $-\frac{\hbar^2}{2m} \nabla^2$, with the familiar spherical Laplacian

$$\nabla^2 = \underbrace{\frac{1}{r^2} \partial_r (r^2 \partial_r)}_{\partial_r^2 + \frac{2}{r} \partial_r} + \underbrace{\frac{1}{r^2 \sin \theta} \partial_\theta (\sin \theta \partial_\theta)}_{\frac{1}{r^2} (\partial_\theta^2 + \cot \theta \partial_\theta)} + \frac{1}{r^2 \sin^2 \theta} \partial_\phi^2.$$

Naive quantization is missing the first-derivative terms $\frac{2}{r} \partial_r$ and $\frac{\cot \theta}{r^2} \partial_\theta$. Where did they go?

Why it fails: $-i\hbar\partial$ is not Hermitian on the curved measure

The physical inner product carries the volume Jacobian:

$$\langle\varphi|\psi\rangle = \int \varphi^* \psi \underbrace{r^2 \sin\theta}_{\text{Jacobian}} dr d\theta d\phi.$$

Is $\hat{p}_r = -i\hbar\partial_r$ Hermitian here? Integrate by parts in r (boundary terms vanish):

$$\int \varphi^* (-i\hbar\partial_r \psi) r^2 dr = \int (-i\hbar\partial_r \varphi)^* \psi r^2 dr - i\hbar \int \varphi^* \psi \underbrace{(\partial_r r^2)}_{=(2/r)r^2} dr,$$

so

$$\boxed{\hat{p}_r^\dagger = -i\hbar\left(\partial_r + \frac{2}{r}\right) \neq \hat{p}_r.}$$

Not Hermitian $\Rightarrow \hat{p}_r^2$ built this way isn't either $\Rightarrow$ no guaranteed real spectrum. The naive recipe **ignored the measure**. (Same for θ : the $\cot\theta$ term.)

The Podolsky / Laplace–Beltrami rule

Cure: read the kinetic energy as a quadratic form, $T = \frac{1}{2}g_{ab}(q)\dot{q}^a\dot{q}^b = \frac{1}{2}g^{ab}(q)p_ap_b$, which defines the metric g_{ab} and the measure $\sqrt{g}d^nq$. The operator Hermitian on $L^2(\sqrt{g}d^nq)$ by construction is

$$\hat{T} = -\frac{\hbar^2}{2} \frac{1}{\sqrt{g}} \partial_a (\sqrt{g} g^{ab} \partial_b), \quad g = \det g_{ab}. \quad (4.5)$$

One integration by parts gives the manifestly symmetric, positive form

$$\int \varphi^* \hat{T} \psi \sqrt{g} d^n q = \frac{\hbar^2}{2} \int (\partial_a \varphi)^* g^{ab} (\partial_b \psi) \sqrt{g} d^n q. \quad (4.6)$$

The $\sqrt{g}$ factors are exactly the measure the naive recipe forgot; for flat Cartesian $g_{ab} = m\delta_{ab}$ it is $-\frac{\hbar^2}{2m} \nabla^2$.

Podolsky recovers the correct Hamiltonian

For the spherical particle, $\mathcal{L}$ gives $g_{ab} = \text{diag}(m, mr^2, mr^2 \sin^2 \theta)$, so

$$\sqrt{g} = m^{3/2} r^2 \sin \theta, \quad g^{ab} = \text{diag}\left(\frac{1}{m}, \frac{1}{mr^2}, \frac{1}{mr^2 \sin^2 \theta}\right).$$

Insert into (4.5); the constant $m^{3/2}$ cancels between $\sqrt{g}$ and $1/\sqrt{g}$:

$$\hat{T} = -\frac{\hbar^2}{2m} \left[\frac{1}{r^2} \partial_r (r^2 \partial_r) + \frac{1}{r^2 \sin \theta} \partial_\theta (\sin \theta \partial_\theta) + \frac{1}{r^2 \sin^2 \theta} \partial_\phi^2 \right] = -\frac{\hbar^2}{2m} \nabla^2. \quad \checkmark$$

The $\frac{2}{r} \partial_r$ and $\cot \theta \partial_\theta$ are **restored automatically** — by the measure, not by hand.

The molecule is the same story

Curvilinear $\{q^A\} = \{\phi, \theta, \chi, Q_1, \dots, Q_f\}$, $T = \frac{1}{2} g^{AB} p_A p_B$, curved measure $\sqrt{g} d^n q$. We now build the three objects — metric g_{AB} , determinant g , inverse g^{AB} — explicitly, then apply (4.5). *Next section.*

Recipe: Podolsky / Laplace–Beltrami quantization

General recipe — curvilinear $\{q^i\}$:

1. **Metric** g_{ij} : read off $T = \frac{1}{2} g_{ij} \dot{q}^i \dot{q}^j$, or
 $g_{ij} = \sum_a m_a (\partial \mathbf{r}_a / \partial q^i) \cdot (\partial \mathbf{r}_a / \partial q^j)$.
2. **Determinant & measure**: $g = \det g_{ij} \Rightarrow$ Hilbert space $L^2(\sqrt{g} d^n q)$.
3. **Inverse** $g^{ij} = (g_{ij})^{-1}$.
4. **Podolsky operator** (Hermitian on $L^2(\sqrt{g} d^n q)$ by construction):

$$\hat{T} = -\frac{\hbar^2}{2} \frac{1}{\sqrt{g}} \partial_i (\sqrt{g} g^{ij} \partial_j);$$

V and coordinate functions $\rightarrow$ multiplication.

5. **(Optional) rescale** $\psi \rightarrow g^{1/4} \psi$ to a flatter measure; leftover c-number = **pseudopotential**.

Instance: free particle in (r, θ, ϕ)

1. $g_{ij} = \text{diag}(m, mr^2, mr^2 \sin^2 \theta)$
2. $\sqrt{g} = m^{3/2} r^2 \sin \theta$
3. g^{ij} : reciprocals of g_{ij}
4. $\hat{T} = -\frac{\hbar^2}{2m} \nabla^2$ ✓
5. flat $\Rightarrow$ no pseudopotential

Molecule (next): metric (4.15) $\rightarrow$ measure (4.21) $\rightarrow$ inverse (4.22) $\rightarrow$ Podolsky $\rightarrow$ Watson $\hat{U}$.

The ro-vibrational geometry

Euler-angle kinematics, explicitly

zyz convention $\mathbf{S} = R_z(\phi)R_y(\theta)R_z(\chi)$ (4.7). Differentiating $\mathbf{S}$ gives $\boldsymbol{\omega} = \mathbf{E}(\Omega)\dot{\Omega}$ and, solving the linear system, $\dot{\Omega} = \mathbf{E}^{-1}\boldsymbol{\omega}$:

$$\mathbf{E} = \begin{pmatrix} -\sin\theta\cos\chi & \sin\chi & 0 \\ \sin\theta\sin\chi & \cos\chi & 0 \\ \cos\theta & 0 & 1 \end{pmatrix}, \quad \mathbf{E}^{-1} = \begin{pmatrix} -\frac{\cos\chi}{\sin\theta} & \frac{\sin\chi}{\sin\theta} & 0 \\ \sin\chi & \cos\chi & 0 \\ \frac{\cos\theta\cos\chi}{\sin\theta} & -\frac{\cos\theta\sin\chi}{\sin\theta} & 1 \end{pmatrix} \quad (4.10, 4.12)$$

with

$$\det \mathbf{E} = -\sin\theta. \quad (4.11)$$

The $1/\sin\theta$ entries are the Euler coordinate singularity at the poles; they are compensated by $|\det \mathbf{E}| = \sin\theta$ in the volume element. (4.10)–(4.12) are the only Euler-angle facts used from here on.

The ro-vibrational metric tensor

The classical BF kinetic energy $2T_{\text{rv}} = \boldsymbol{\omega}^T \mathbf{I}_N \boldsymbol{\omega} + 2\boldsymbol{\omega}^T \sum_b \boldsymbol{\sigma}_b \dot{Q}_b + \sum_b \dot{Q}_b^2$ (4.13), re-expressed in the *coordinate* velocities via $\boldsymbol{\omega} = \mathbf{E}\dot{\boldsymbol{\Omega}}$, defines the metric:

$$g_{AB} = \begin{pmatrix} \mathbf{E}^T \mathbf{I}_N \mathbf{E} & \mathbf{E}^T \boldsymbol{\sigma} \\ \boldsymbol{\sigma}^T \mathbf{E} & \mathbb{1}_f \end{pmatrix} \quad (4.15)$$

- Euler–Euler block: inertia pulled back through the kinematic matrix.
- Euler–mode block: Coriolis coupling $\boldsymbol{\sigma}_b = \sum_a \zeta_{ab} Q_a$ (constant ζ 's).
- mode–mode block: *identity* — the gift of mass-weighted rectilinear normal coordinates.

Every entry is an explicit function of (θ, χ, Q) .

Electrons: flat, constant, decoupled from the measure

Including electron velocities, the full mass matrix has a *constant* electron block D. Eliminating it (Schur complement = completing the square) removes all ω -dependence:

$$CD^{-1}C^T = \mathbf{I}_e - \mathbf{I}_X \implies \det M = \underbrace{\det D}_{\text{const}} \cdot \det \begin{pmatrix} \mathbf{I}_N & \boldsymbol{\sigma} \\ \boldsymbol{\sigma}^T & \mathbb{1}_f \end{pmatrix} \quad (4.17)$$

— the same $\mathbf{I}_e/\mathbf{I}_X$ cancellation as the classical cross-check of § 3.6.

- Electron measure is the flat $\prod_i d^3\bar{\mathbf{r}}_i$; $\hat{\mathbf{p}}_i = -i\hbar\nabla_{\bar{\mathbf{r}}_i}$ is Hermitian on it.
- $\hat{L}_{\text{elec},\alpha}$ is Hermitian *without symmetrization*: its vector field is divergence-free, $\sum_{i\gamma} \partial_{\bar{r}_{i\gamma}} (\epsilon_{\alpha\beta\gamma} \bar{r}_{i\beta}) = 0$ (4.18).

The entire quantization problem lives in the ro-vibrational sector; electrons enter only via the flat kinetic operator and the shift $\hat{J}_\alpha \rightarrow \hat{J}_\alpha - \hat{L}_{\text{elec},\alpha}$.

Determinant and volume element

Block row reduction gives $\det \begin{pmatrix} \mathbf{a} & \mathbf{b} \\ \mathbf{b}^T & \mathbb{1} \end{pmatrix} = \det(\mathbf{a} - \mathbf{b}\mathbf{b}^T)$ (4.19). Applied to (4.15):

$$\det g_{AB} = \sin^2 \theta \gamma(Q), \quad \gamma \equiv \det \mathbf{I}', \quad \mathbf{I}' \equiv \mathbf{I}_N - \boldsymbol{\sigma}\boldsymbol{\sigma}^T. \quad (4.20)$$

The Watson-modified inertia appears by itself

The determinant is governed by $\mathbf{I}' = \mathbf{I}_N - \boldsymbol{\sigma}\boldsymbol{\sigma}^T$ — *not* the bare $\mathbf{I}_N$: the Coriolis coupling reduces the rotational determinant. It is the same $\mathbf{I}'$ whose inverse $\boldsymbol{\mu}$ is the classical rotational kernel (3.20).

$$\sqrt{g} d^{3+f} q = \underbrace{\sin \theta d\phi d\theta d\chi}_{\text{Haar measure on } SO(3)} \cdot \underbrace{\sqrt{\gamma(Q)} \prod_b dQ_b}_{\text{rot.-vib. Jacobian}} \cdot \underbrace{\prod_i d^3 \mathbf{r}_i}_{\text{flat}} \quad (4.21)$$

The inverse metric

Claim, then *verify* by block multiplication using only $\mu \mathbf{l}' = \mathbb{1}$ [(4.23)]:

$$g^{AB} = \begin{pmatrix} E^{-1} \mu E^{-T} & -E^{-1} \mu \sigma \\ -\sigma^T \mu E^{-T} & \mathbb{1}_f + \sigma^T \mu \sigma \end{pmatrix}, \quad \mu = (\mathbf{I}_N - \sigma \sigma^T)^{-1}. \quad (4.22)$$

Two inversions, one computation

The Watson reciprocal inertia μ — introduced classically by inverting the momentum–velocity relation (3.20) — reappears as the geometric inverse of the metric.

e.g. top-left: $E^T \mathbf{I}_N E \cdot E^{-1} \mu E^{-T} - E^T \sigma \cdot \sigma^T \mu E^{-T} = E^T (\mathbf{I}_N - \sigma \sigma^T) \mu E^{-T} = \mathbb{1}_3$. $\checkmark$

Quantizing rotation & Coriolis coupling

Why the $\hat{J}_\alpha$ look unlike the usual angular momentum

The rotational operators below differ from the $\hat{L}$ of introductory QM for *three* reasons:

1. **Three angles, not two.** A rigid body needs (ϕ, θ, χ) , so besides $\partial_\phi, \partial_\theta$ there is a third generator $\hat{J}_3 = -i\hbar\partial_\chi$, and the three mix.
2. **Body-fixed, not space-fixed.** They are components along the *molecule's* axes, $\hat{J}_\alpha^{\text{BF}} = S_{\beta\alpha}(\Omega)\hat{J}_\beta^{\text{SF}}$; since $\mathbf{S}$ depends on the angles, their commutator carries the **anomalous sign** $[\hat{J}_\alpha, \hat{J}_\beta] = -i\hbar\epsilon_{\alpha\beta\gamma}\hat{J}_\gamma$ — which we *never use*.
3. **Convention-dependent form.** The explicit trig coefficients are fixed by our *zyz* choice $\mathbf{S} = R_z(\phi)R_y(\theta)R_z(\chi)$; another convention relabels them (*zyz* is itself standard — Wigner, Zare).

Nothing is postulated — the operators fall out of the Laplace–Beltrami expansion next.

Laplace–Beltrami on the rotational block $\Rightarrow \hat{J}_\alpha$

Put the measure $\sqrt{g} = \sin \theta \sqrt{\gamma}$ and the rotational inverse-metric block $g_{st}^{\Omega\Omega} = (E^{-1})_{s\alpha} \mu_{\alpha\beta} (E^{-1})_{t\beta}$ into the Podolsky/LB operator (4.5). Since $\mu, \sqrt{\gamma}$ depend on Q only, they pass through ∂_Ω :

$$\hat{T}_{\text{rot}} = -\frac{\hbar^2}{2} \mu_{\alpha\beta} \frac{1}{\sin \theta} \partial_s (\sin \theta (E^{-1})_{s\alpha} (E^{-1})_{t\beta} \partial_t).$$

The combination $(E^{-1})_{s\alpha} \partial_{\Omega_s}$ appears twice — name it:

$$\boxed{\begin{aligned} \hat{J}_1 &= -i\hbar \left[-\frac{\cos \chi}{\sin \theta} \partial_\phi + \sin \chi \partial_\theta + \frac{\cos \theta \cos \chi}{\sin \theta} \partial_\chi \right], \\ \hat{J}_\alpha \equiv -i\hbar \sum_s (E^{-1})_{s\alpha} \partial_{\Omega_s} : \quad \hat{J}_2 &= -i\hbar \left[\frac{\sin \chi}{\sin \theta} \partial_\phi + \cos \chi \partial_\theta - \frac{\cos \theta \sin \chi}{\sin \theta} \partial_\chi \right], \\ \hat{J}_3 &= -i\hbar \partial_\chi, \end{aligned}} \quad (4.24)$$

read row-by-row off E^{-1} (4.12). One leftover term $\partial_s (\sin \theta E^{-1})$ remains — next.

The divergence identity makes it clean

Expanding the outer derivative gives the clean double-derivative *plus* a first-order remainder:

$$\hat{T}_{\text{rot}} = \frac{1}{2}\mu_{\alpha\beta}\hat{J}_{\alpha}\hat{J}_{\beta} - \underbrace{\frac{\hbar^2}{2}\mu_{\alpha\beta}\left[\frac{1}{\sin\theta}\partial_s(\sin\theta(E^{-1})_{s\alpha})\right]}_{\text{divergence (4.26)}}(E^{-1})_{t\beta}\partial_t.$$

The divergence vanishes term by term:

$$\alpha=1 : \partial_{\phi}(-\cos\chi) + \partial_{\theta}(\sin\theta\sin\chi) + \partial_{\chi}(\cos\theta\cos\chi) = 0,$$

$$\alpha=2 : \partial_{\phi}(\sin\chi) + \partial_{\theta}(\sin\theta\cos\chi) + \partial_{\chi}(-\cos\theta\sin\chi) = 0, \quad \alpha=3 : \partial_{\chi}\sin\theta = 0.$$

Why the rotation block quantizes so cleanly (for the advanced reader)

Podolsky's rule *should* spawn $\partial\sqrt{g}$ and $\partial g^{\Omega\Omega}$ first-order terms — but they collapse into this one divergence, zero because the rotation generators are divergence-free on $\sin\theta d\Omega$. Hence $\hat{T}_{\text{rot}} = \frac{1}{2}\mu_{\alpha\beta}\hat{J}_{\alpha}\hat{J}_{\beta}$, Hermitian, **no pseudopotential** (4.29).

The Coriolis (rotation–vibration) block — also clean

The off-diagonal blocks $g^{\Omega Q} = -E^{-1} \mu \sigma$ give the cross term (4.30):

$$\hat{T}_{\text{Cor}} = -\frac{1}{2} \left[\hat{J}_\alpha \mu_{\alpha\beta} \hat{\pi}_\beta + \frac{1}{\sqrt{\gamma}} \hat{P}_b \sqrt{\gamma} \sigma_{\alpha b} \mu_{\alpha\beta} \hat{J}_\beta \right], \quad \hat{\pi}_\alpha = \sum_b \sigma_{\alpha b}(Q) \hat{P}_b.$$

The two groups are **adjoints of each other** on the measure (4.21) — the $\gamma^{\pm 1/2}$ are exactly what integration by parts over $\sqrt{\gamma} dQ$ produces — so their sum is Hermitian *automatically*, with no residue.

Section result

Rotation (4.29) and Coriolis (4.30) quantize with **no** pseudopotential — the divergence identity and the adjoint-pair structure do all the work. The flat-measure rescaling of the next section leaves both untouched (giving the symmetric $-\frac{1}{2}(\hat{J}\mu\hat{\pi} + \hat{\pi}\mu\hat{J})$, 4.35). **Every pseudopotential comes from the vibrational block.**

The vibrational block & the Watson pseudopotential

The vibrational block is different

The vibrational inverse-metric block is $g^{QQ} = \mathbb{1} + \sigma^T \mu \sigma$; the LB operator on the curved measure $\sqrt{\gamma}$ is

$$\hat{T}_{\text{vib}} = -\frac{\hbar^2}{2} \frac{1}{\sqrt{\gamma}} \partial_{Q_b} (\sqrt{\gamma} g_{bc}^{QQ} \partial_{Q_c}).$$

Expand the δ_{bc} part by one product rule:

$$-\frac{\hbar^2}{2} \frac{1}{\sqrt{\gamma}} \partial_{Q_b} (\sqrt{\gamma} \partial_{Q_b} \psi) = \frac{1}{2} \hat{P}_b^2 \psi - \underbrace{\frac{\hbar^2}{2} \left(\frac{1}{2} \partial_{Q_b} \ln \gamma \right) \partial_{Q_b} \psi}_{\text{first-order remainder}}.$$

Unlike rotation, this remainder does not vanish: $\partial_{Q_b} \ln \gamma \neq 0$ because $\gamma = \det \mathbf{I}'(Q)$ genuinely depends on the shape, and the bare $\hat{P}_b$ is *not* Hermitian on $\sqrt{\gamma} dQ$. How to handle it?

Warm-up: the radial part of the spherical Laplacian

Read the radial coordinate as an effective one-body *vibration* (a diatomic bond). Its Laplace–Beltrami operator, on the curved measure $r^2 dr$, is

$$\hat{T}_r = -\frac{\hbar^2}{2m} \frac{1}{r^2} \partial_r (r^2 \partial_r), \quad \int |R|^2 r^2 dr = 1.$$

But vibrational wavefunctions are conventionally normalized on a *flat* measure. Rescale $u = rR$ (i.e. $\psi \rightarrow g^{1/4} \psi$, $\sqrt{g} \propto r^2$) so $\int |u|^2 dr = 1$:

$$\hat{T}_r \xrightarrow{u=rR} -\frac{\hbar^2}{2m} \frac{d^2}{dr^2} \text{ on the flat measure } dr.$$

The awkward $\frac{2}{r} \partial_r$ is gone — absorbed into the rescaling. *This* move is what, in general, leaves a pseudopotential behind.

Two schemes: Laplace–Beltrami vs Podolsky

The *same* operator, in two representations:

Laplace–Beltrami

wavefn on curved $\sqrt{g} d^n q$

$$-\frac{\hbar^2}{2\sqrt{g}} \partial(\sqrt{g} g^{ij} \partial)$$

no explicit potential

Podolsky (flat measure)

$\psi_W = g^{1/4} \psi$ on flat $d^n q$

$$\frac{1}{2} \hat{p}_i g^{ij} \hat{p}_j + U$$

extra c-number $U =$ pseudopotential

Related by conjugation with $g^{1/4}$; U is the bookkeeping cost of the flat measure.

Radial case: $U = 0$

For $\gamma \propto r^2$ the residual cancels ($u = rR$ is exact) — like the rigid rotor. A genuine shape-dependent $\gamma(Q)$ does **not** cancel: that is where the Watson potential is born.

Rescaling the molecular vibrational block

Same move with $\gamma = \det \mathbf{I}'(Q)$: absorb the shape factor, $\psi_W = \gamma^{1/4} \psi$, so the vibrational measure is flat, $\prod_b dQ_b$ (normal-mode space treated as flat). Conjugating $\hat{T}_{\text{vib}}$ (one product rule per line):

$$\begin{aligned}\partial_{Q_b}(\gamma^{-1/4} \psi_W) &= \gamma^{-1/4} [\partial_{Q_b} \psi_W - \tfrac{1}{4}(\partial_{Q_b} \ln \gamma) \psi_W], \\ \partial_{Q_b}(\gamma^{1/4} [\cdots]) &= \gamma^{1/4} [\partial_{Q_b}^2 \psi_W - \tfrac{1}{16}(\partial_{Q_b} \ln \gamma)^2 \psi_W - \tfrac{1}{4}(\partial_{Q_b}^2 \ln \gamma) \psi_W].\end{aligned}$$

The cross terms cancel; the bare $\frac{1}{2} \sum_b \hat{P}_b^2$ survives *plus* a multiplicative residue

$$U_{\text{vib}} = \frac{\hbar^2}{8} \sum_b \left[\partial_{Q_b}^2 \ln \gamma + \tfrac{1}{4}(\partial_{Q_b} \ln \gamma)^2 \right] \quad (+ U_{\text{Cor}} \text{ from the } \boldsymbol{\sigma}^T \boldsymbol{\mu} \boldsymbol{\sigma} \text{ dressing}).$$

Now the residue is nonzero — the promised pseudopotential.

Collapse to the Watson trace

The residue $U_{\text{vib}} + U_{\text{Cor}}$ collapses through two exact Q -calculus identities. **Jacobi:**

$\partial_{Q_b} \ln \gamma = \text{tr}(\boldsymbol{\mu} \partial_{Q_b} \mathbf{I}')$. **Watson's closed form (4.37):** $\mathbf{I}' = A(\mathbf{I}^e)^{-1}A$ with

$A = \mathbf{I}^e + \frac{1}{2} \sum_b a^b Q_b$, so every derivative reduces to constants; the Eckart sum rules fix the remainder:

$$\hat{U}(Q) = -\frac{\hbar^2}{8} \sum_{\alpha=1}^3 \mu_{\alpha\alpha}(Q). \quad (4.38)$$

- A pure c -number from the *measure* — kernel ordering never makes one.
- Electrons don't modify it ($\gamma, \boldsymbol{\mu}$ depend on Q alone).
- Special to *rectilinear* normal coordinates (Watson's closed form).

The complete quantum Hamiltonian

Electronic kinetic operator

In BF Cartesian electron coordinates $\hat{\mathbf{p}}_i = -i\hbar\nabla_{\mathbf{r}_i}$ is already Hermitian on the flat measure, so the electron + mass-polarization piece quantizes directly:

$$\hat{T}_e = \sum_i \frac{\hat{\mathbf{p}}_i^2}{2m_e} + \frac{1}{2M_N} \left(\sum_i \hat{\mathbf{p}}_i \right)^2 = \sum_i \frac{\hat{\mathbf{p}}_i^2}{2\mu_e} + \frac{1}{M_N} \sum_{i<l} \hat{\mathbf{p}}_i \cdot \hat{\mathbf{p}}_l. \quad (4.39, 4.40)$$

The diagonal part renormalizes the electron mass, $\frac{1}{m_e} + \frac{1}{M_N} = \frac{1}{\mu_e}$; the off-diagonal two-electron term is the mass polarization *sensu stricto* (nuclear recoil).

The exact non-relativistic molecular Hamiltonian

Collecting the flat-measure blocks — rotational (4.29), Coriolis (4.35), vibrational, pseudopotential (4.38), electronic (4.39), and the shift $\hat{J}_\alpha \rightarrow \hat{J}_\alpha - \hat{L}_{\text{elec},\alpha}$:

$$\begin{aligned}
 \hat{H}_{\text{int}} = & \frac{1}{2} \sum_{\alpha\beta} (\hat{J}_\alpha - \hat{L}_{\text{elec},\alpha} - \hat{\pi}_\alpha) \mu_{\alpha\beta}(Q) (\hat{J}_\beta - \hat{L}_{\text{elec},\beta} - \hat{\pi}_\beta) && \text{rotation-vibration-electron} \\
 & + \frac{1}{2} \sum_b \hat{P}_b^2 && \text{vibration} \\
 & + \sum_i \frac{\hat{\mathbf{p}}_i^2}{2m_e} + \frac{1}{2M_N} \left(\sum_i \hat{\mathbf{p}}_i \right)^2 && \text{electrons + mass pol.} \\
 & - \frac{\hbar^2}{8} \sum_\alpha \mu_{\alpha\alpha}(Q) && \text{Watson pseudopotential} \\
 & + V_{NN} + V_{Ne} + V_{ee} && \text{Coulomb}
 \end{aligned} \tag{4.41}$$

on $L^2(d\mu_W)$, $d\mu_W = \sin\theta d\phi d\theta d\chi \prod_b dQ_b \prod_i d^3\mathbf{r}_i$. Every operator is an explicit differential operator, Hermitian by integration by parts.

Summary

The journey, in four stages

1. **Classical**: lab H (1.1) $\rightarrow$ one mixed mass-centered point transformation $\rightarrow H$ (2.12): COM exactly split for any $\mathbf{P}_{\text{cm}}$, mass polarization included — no boost, no rest frame.
2. **Body-Fixed**: velocity-form Lagrangian $(\mathbf{I}_N + \mathbf{I}_e - \mathbf{I}_X)$ $\rightarrow$ key identity $\mathbf{J} - \mathbf{L}_{\text{elec}} = \mathbf{I}_N \boldsymbol{\omega} + \mathbf{L}_N^{\text{vib}}$ ($\mathbf{I}_e, \mathbf{I}_X$ cancel) $\rightarrow$ classical H_{int} (3.22) with Watson $\boldsymbol{\mu} = (\mathbf{I}_N - \boldsymbol{\sigma} \boldsymbol{\sigma}^T)^{-1}$.
3. **Quantize**, coordinate-natively: metric (4.15) $\rightarrow$ measure (4.21) $\rightarrow$ inverse (4.22) $\rightarrow$ Podolsky blocks $\rightarrow \gamma^{1/4}$ rescaling $\rightarrow$ Watson $\hat{U} = -\frac{\hbar^2}{8} \sum_{\alpha} \mu_{\alpha\alpha}$ (4.38). Hermiticity by integration by parts + divergence identity (4.26) — **no angular-momentum algebra**.
4. **Quantum $\hat{H}$** : the exact non-relativistic $\hat{H}_{\text{int}}$ (4.41).

Lagrangians & Hamiltonians: stages 1-2

1. Lab-frame Cartesian.

$$\mathcal{L} = \sum_j \frac{1}{2} M_j |\dot{\mathbf{R}}_j|^2 + \sum_i \frac{1}{2} m_e |\dot{\mathbf{r}}_i|^2 - V, \quad H = \sum_j \frac{\mathbf{P}_j^2}{2M_j} + \sum_i \frac{\mathbf{p}_i^2}{2m_e} + V.$$

2. Mass-centered (total CoM $\mathbf{R}_{\text{cm}}$; internals referred to the nuclear CoM):

$$\mathcal{L} = \frac{1}{2} M_{\text{tot}} |\dot{\mathbf{R}}_{\text{cm}}|^2 + \sum_j \frac{1}{2} M_j |\dot{\mathbf{R}}'_j|^2 + \sum_i \frac{1}{2} m_e |\dot{\mathbf{r}}'_i|^2 - \frac{m_e^2}{2M_{\text{tot}}} \left(\sum_i \dot{\mathbf{r}}'_i \right)^2 - V,$$

$$H = \frac{\mathbf{P}_{\text{cm}}^2}{2M_{\text{tot}}} + \sum_j \frac{\mathbf{P}'_j{}^2}{2M_j} + \sum_i \frac{\mathbf{p}'_i{}^2}{2m_e} + \frac{1}{2M_N} \left(\sum_i \mathbf{p}'_i \right)^2 + V.$$

Mass polarization: velocity form $-\frac{m_e^2}{2M_{\text{tot}}} (\sum \dot{\mathbf{r}}')^2 \leftrightarrow$ momentum form $+\frac{1}{2M_N} (\sum \mathbf{p}')^2$.

Lagrangians & Hamiltonians: stages 3–4

3. Body-fixed (Eckart) Cartesian — electrons in velocity form:

$$\mathcal{L} = \frac{1}{2} M_{\text{tot}} |\dot{\mathbf{R}}_{\text{cm}}|^2 + \frac{1}{2} \boldsymbol{\omega}^T (\mathbf{I}_N + \mathbf{I}_e - \mathbf{I}_X) \boldsymbol{\omega} + \boldsymbol{\omega} \cdot (\mathbf{L}_N^{\text{vib}} + \mathbf{L}_e - \mathbf{L}_X) \\ + \frac{1}{2} \sum_j M_j |\dot{\mathbf{R}}_j|^2 + \frac{m_e}{2} \sum_i |\dot{\mathbf{r}}_i|^2 - \frac{m_e^2}{2M_{\text{tot}}} |\dot{\mathbf{r}}|^2 - V,$$

$$H = \frac{\mathbf{P}_{\text{cm}}^2}{2M_{\text{tot}}} + \sum_j \frac{\bar{\mathbf{P}}_j^2}{2M_j} + \sum_i \frac{\bar{\mathbf{p}}_i^2}{2m_e} + \frac{1}{2M_N} \left(\sum_i \bar{\mathbf{p}}_i \right)^2 + V.$$

4. Normal-mode (Watson): $\mathcal{L}$ as in stage 3 with $\frac{1}{2} \sum_j M_j |\dot{\mathbf{R}}_j|^2 \rightarrow \frac{1}{2} \sum_b \dot{Q}_b^2$; the canonical Hamiltonian is

$$H_{\text{int}} = \frac{1}{2} (\mathbf{J} - \mathbf{L}_{\text{elec}} - \boldsymbol{\pi})^T \boldsymbol{\mu} (\mathbf{J} - \mathbf{L}_{\text{elec}} - \boldsymbol{\pi}) + \frac{1}{2} \sum_b P_b^2 + \sum_i \frac{\bar{\mathbf{p}}_i^2}{2m_e} + \frac{1}{2M_N} \left(\sum_i \bar{\mathbf{p}}_i \right)^2 + V_{NN} + V_{Ne} + V_{ee}.$$

CoM decouples throughout ($\mathbf{P}_{\text{cm}}^2/2M_{\text{tot}}$); quantizing stage 4 gives $\hat{H}_{\text{int}}$ (4.41).

What we did — and did not — assume

$\hat{H}_{\text{int}}$ (4.41) is **exact**. The only inputs were:

- non-relativistic dynamics;
- the Eckart frame (a convention);
- $3N_{\text{nuc}} - 6$ normal coordinates (an exact parametrization).

No Born–Oppenheimer, no clamped nuclei, no neglect of mass polarization, no harmonic assumption — and **no anomalous commutator as input**.

**The full molecular Hamiltonian:
exact, quantized coordinate-natively,
and ready to approximate.**